Internet Booking Engine

Brief

- A **multi-tenant internet booking engine** that can be **configured based on the tenant's requirements.**
- Users should be able to view available properties, search for rooms, and complete reservations seamlessly.
- The system supports promotions, allowing users to choose discounts.
- Users can share search results easily and manage their bookings even through a guest user workflow.
- Two different environments are expected to be provisioned and maintained.
    - "DEV" environment for development and testing
    - "PROD" environment for a 24x7 stable deployed version every sprint-end.

UX & Reference Links

To better understand the user interactions and expected UI/UX, refer to the following resources:

- **Figma Wireframes**: Here is the Figma link for the wireframes. This provides a structured view of the user experience and design flow.
- **Reference Website**: Play around with this website to understand the user interactions better and get a feel for the booking process.

Functionality

1. Search

a. Support multiple rooms for a single room type.
b. If the search is performed from the landing page, ensure that configurable fields have default values for consistency while fetching results.
c. Consider room availability across the entire length of stay when searching.
d. Store users' previous search results and repopulate them if they revisit the room results page.
e. Navigating from the landing page to the room results page should generate a shareable URL with search parameters as query parameters, ensuring consistency in results.
f. Follow this precedence for search parameters while fetching results:
    i. **P0** - Query parameters
    ii. **P1** - Previous search inputs

2. Guest User Workflow

a. Allow guest users to search for rooms, view details, and proceed to checkout without requiring an account.
b. Ensure guest users can easily share their search results and booking details with others.
c. Maintain session persistence so guest users can navigate freely without losing their progress.

3. User Login Workflow (Bonus)

a. Provide login and sign-up functionality using AWS Cognito and AWS Amplify.
b. Ensure users can reset their passwords securely.

4. Configurability

The following fields and components should be configurable:

a. **Landing Page**
    i. Header logo
    ii. Page title
    iii. Banner image
    iv. Length of stay
    v. Guest, room, and accessibility options in the search form
    vi. Guest dropdown options
    vii. Room dropdown options
b. **Room Results Page**
    i. Filters
    ii. Filter options
    iii. Sort options
c. **Room Details Modal**
    i. Room type images
    ii. Room type description
    iii. Room type amenities
d. **Checkout Page**
    i. Traveler info fields
    ii. Billing info fields
    iii. Taxes and surcharges
    iv. % Due at resort

5. Rate Calendar

a. Displays the minimum nightly rate for a particular day across all room-types.
b. The label next to the “Apply Dates” button should display the minimum nightly rate across the entire length of stay.

6. Room Results

a. **Filters** - Define how new filters will be evaluated, given that filters are configurable.

b. **Pagination** - Decide whether pagination will be handled on the UI only or if the API will also implement pagination.

7. Checkout Page

a. Ensure all fields are validated before processing a booking request:
   i. Email addresses
   ii. Phone numbers
   iii. Zip codes (support 5-digit and 6-digit formats)
   iv. Credit card numbers (support both 15-digit and 16-digit formats)
   v. Expiry dates (cannot be in the past)
   vi. CVV (3 or 4 digits)
   vii. Valid states for the selected country
   viii. Ensure the backend recalculates the due amount instead of relying on the frontend value.

8. Booking

a. Handle concurrent bookings incrementally:
   i. Establish concurrency at the room type level.
   ii. Manage concurrency across mutually exclusive date ranges.
   iii. Manage concurrency for overlapping dates within the same room type.
   iv. Assign rooms to a booking dynamically if the same room is not available for the entire stay.
   v. Generate a unique URL for each successful booking, allowing guest users to access it anytime.
   vi. Share the unique booking URL with users upon confirmation.

9. My Bookings

a. Navigated from the “My Bookings” button visible on the header.
b. Allows viewing and management of all bookings for a **logged-in** user.

UI Requirements

1. Header and Footer Requirements

The header and footer should be consistent across all pages.

Header

The header should include the following elements:

- **Logo and Booking Engine Name**: Clicking either the logo or the name should redirect users to the home page.
- **Login/Logout Button**:
  - If a user is logged in, display a **Logout** button.

- If no user is logged in, display a **Login** button.
- Logged-in users will have access to the “My Bookings” page. Toggle the visibility of this button accordingly.

Footer

The footer should include:

1. **Company Logo**
2. **Copyright Information**: Clearly state the copyright details regarding the usage of content from the website.

2. Landing Page Requirements

The landing page is the first page users see when they visit the website. It should include the following elements:

a. Banner Image

A visually appealing banner image should be displayed at the top of the page.

b. Search Form

The search form allows users to filter their search results. The fields within the search form must be **configurable**, meaning the website owner can choose which fields to display. The following fields are expected:

Fixed Fields

These fields must always be present in the search form:

- **Property Name Field**
  - A dropdown listing the properties owned by the tenant.
  - For this project, each team will work with only one assigned property. ?
- **Booking Date Range**
  - Clicking this field opens a rate calendar where users select their stay’s start and end dates.
  - The calendar displays **two months at a time** (current month and next month), with navigation arrows to browse further months.
  - Below each date, the **minimum nightly rate** across all rooms for that property is displayed.
  - **"Apply Dates" Button**:
    - Initially disabled until both start and end dates are selected.

        - Once dates are selected, the button becomes active, and a label shows the **minimum nightly rate** for the selected range.
    - **Date Selection Rules**:
        - If only the start date is selected, the end date can only be chosen within the **configured length of stay** set by the owner.
        - Past dates cannot be selected as the start date.
        - All dates before today must be disabled and should not display nightly rates.
        - A message should appear below the **"Apply Dates"** button informing users of the configured length of stay.

Configurable Fields

These fields are optional and can be enabled or disabled based on the website owner's preferences:

- **Number of Guests**
    - If enabled, clicking this field opens a dropdown for users to select the number of guests for each guest type (e.g., Adults, Teens, Children).
    - Guest types and their availability are **configurable by the owner**, meaning options may vary by tenant.
    - There should be a **maximum limit** on the number of guests.
    - If this field is disabled, a **default value** should be assumed.
    - Negative guest counts are not allowed.
    - Once the dropdown is closed, a summary of the selection should be displayed (e.g., "3 Adults, 1 Child").
- **Number of Rooms**
    - If enabled, a dropdown allows users to select the number of rooms.
    - The dropdown options should be **configurable**.
    - If disabled, a **default value** should be assumed.
- **Wheelchair Accessibility Checkbox**
    - If enabled, users can specify whether they require wheelchair-accessible rooms.
    - If disabled, a **default value** should be assumed.

c. Search Button

- If all required fields in the search form are filled, clicking the **Search** button should redirect users to the **room results page**, passing the search parameters as **query parameters in the URL**.
- The button should remain **disabled** until all required fields are populated.

3. Room Results Page

The Room Results page displays details about available rooms based on the selected date range and search parameters. It includes the following components:

a. Banner

- A simple banner at the top displaying an image of the property.

b. Stepper

- A stepper to **visualize and navigate** through different steps of the booking process.
- **Step Completion Rules**:
  - Completed steps are marked with a **checkmark**.
  - Users **cannot proceed** to a future step without completing the current step.
  - Users **can navigate back** to any previous step at any time by clicking on the "stepper circles".
  - The line joining the "stepper circles" is blue when you go to the next stage.
  - When user lands on checkout page, they have an option to choose "Continue Shopping" and "Cancel"
    - Clicking on "Continue Shopping" will redirect the user to the room results page but the checkout context will be preserved.
      - The "stepper circle" moves to step-2 but step-3 is clickable (redirects them again to this checkout page). The line joining the last two "stepper circles" stays blue.
    - Clicking on "Cancel" will redirect the user to the room results page but the checkout context will be cleared.
      - The "stepper circle" moves to step-2 and step-3 is un-clickable. The line joining the last two "stepper circles" turns back to grey.

c. Search Form

- A horizontal search form with the **same fields as the Landing Page search form**.
- Fields that were hidden on the Landing Page will now be displayed with their **default values**.
- Users can update search results by modifying the values in this form.
- Clicking the **Search** button updates the room results based on the new search parameters.

4. Room Results Panel

This panel displays details about the available rooms based on the user's search criteria. It consists of the following components:

Room Cards

Each room card represents a type of room available and contains:

- **Image Carousel**
    - A carousel displaying images of the room type.
    - Left and right arrows allow **manual navigation** between images.
- **Room Details**
    - Room size (in sq. ft.)
    - Number of beds
    - Maximum occupancy
- **Room Title and Description**
- **Total Price for the stay**
    - The calculated total price for the selected date range.
- **"Select Room" Button**
    - Clicking this button **opens a modal** with more detailed information about the selected room.

Room Result Filters

- A list of filter options to refine search results.
- **Configurable by the tenant** (the owner decides which filters and filter options to display).
- Selecting a filter option **automatically updates** the room results.

Pagination

- Results should be **paginated** to improve performance and usability.
- A label at the top of the panel should indicate:
    - The **number of results displayed**
    - The **total number of results available**

Sorting

- A sorting feature allows users to arrange room results based on **predefined criteria**.
- The available sorting options are **configurable by the tenant**.
- Users should be able to **toggle between ascending and descending order**.

5. Room Details Modal Requirements

When a user clicks the **"Select Room"** button on a room card, this modal appears, providing detailed information about the selected room.

1. Room Image Carousel

- A **carousel** displaying images of the room type.
  - Users can manually navigate using **left and right arrows**.

2. Room Information

- **Room Type Title**
- **Room Details** including:
  - **Occupancy** (maximum number of guests allowed)
  - **Number of Beds**
  - **Room Area** (in sq. ft.)
  - **Amenities** available in the room
  - **Description** of the room type

3. Pricing Options

Room prices are displayed in **two sections** based on the applicable rates:

a. Standard Rates

- Displays the **standard cost** of the room without any deals or discounts.
- Shows the **total price** over the selected length of stay.
- A **"Select Package" button** allows the user to choose this option and proceed to the **Checkout Page**.

b. Deals & Packages

- A list of **available deals** based on booking dates and guest details.
- Each deal is displayed in a **card format**, containing:
  1. **Deal Title**
  2. **Deal Description**
  3. **Discounted Price for the stay** (after applying the deal)
  4. **"Select Package" Button** (choosing this redirects the user to the **Checkout Page**)

5. Modal Closing Options

Users can close the modal by:

- Clicking **outside** the modal

- Clicking the **“X” button** at the top right corner

6. Checkout Page Requirements

The **Checkout Page** collects traveler details, billing information, and payment details to finalize the booking. Users land on this page after selecting their package.

1. Booking Information Panel

This panel is divided into **three sections**, and users must complete each section before proceeding to the next.

a. Traveler Information Section

This section collects traveler details.

- **Required Fields:**
  - **First Name** (Required)
  - **Last Name** (Optional)
  - **Phone Number** (Required, must be validated)
  - **Email ID** (Required, must be validated)
- If any field contains **invalid data**, an error message appears below it in **red**.
- **Navigation Buttons:**
  - **"Next: Billing Info" Button:**
    - Becomes **active only when all required fields are valid**.
    - Clicking it collapses the **Traveler Info** section and expands the **Billing Info** section.

b. Billing Information Section

This section collects billing details.

- **Required Fields:**
  - **First Name**
  - **Mailing Address 1**
  - **Country** (Dropdown, must be validated)
  - **City**
  - **State** (Dropdown, must be a valid option for the selected country)
  - **ZIP Code** (Must be a valid format for the selected state)
  - **Phone Number** (Must be validated)
  - **Email ID** (Must be validated)
- **Optional Fields:**
  - **Last Name**

    - **Mailing Address 2**
- If any field contains **invalid data**, an error message appears below it in **red**.
- **Navigation Buttons:**
    - **"Next: Payment Info" Button:**
        - Becomes **active only when all required fields are valid**.
        - Clicking it collapses the **Billing Info** section and expands the **Payment Info** section.
    - **"Previous: Traveler Info" Button:**
        - Clicking it collapses the **Billing Info** section and reopens the **Traveler Info** section.

c. Payment Information Section

This section collects payment details.

- **Required Fields:**
    - **Card Number** (Masked & validated)
    - **Expiry Month** (Validated)
    - **Expiry Year** (Validated)
    - **CVV Code** (Masked & validated)
- If any field contains **invalid data**, an error message appears below it in **red**.
- **Additional Options:**
    - **Checkbox (Required):** Must agree to the terms and conditions.
        - Clicking on **"Terms and Conditions"** opens a modal to show the **Terms & Conditions**.
- **Amount Due:** Displayed before completing the booking.
- **Navigation Buttons:**
    - **"Previous: Billing Info" Button:**
        - Clicking it collapses the **Payment Info** section and reopens the **Billing Info** section.
    - **"Complete Booking" Button:**
        - Finalizes the booking process.

2. Trip Itinerary Panel

It provides a breakdown of the booking cost and additional details.

a. Cost Breakdown

- Displays a summary of the **entire booking cost**, including:
    1. Applicable **taxes**
    2. Amount **to be paid now** and **to be paid at the resort**
    3. Any applied **promo codes**

b. Info Icons

- Clicking the **info icon** next to **taxes and surcharges** opens a modal with a detailed breakdown of the applied rates.
- Clicking the **info icon** next to **promotions applied** opens a modal explaining the applied promotions.

c. "Continue Shopping" Button

- A button at the bottom of the Trip Itinerary Panel labeled **"Continue Shopping"**.
- Clicking this button redirects users to the **Checkout Page** but the checkout context will be preserved.
    - Users can come back to this page by clicking the stepper's Step:3.

3. Contact Details Section

- Displays **customer support contact number** and **help desk hours**.

7. Booking Confirmation Page

The **Booking Confirmation Page** is displayed after a successful booking and the booking details are mailed to their **email address**.

This page provides details of the booking.

1. Booking Overview

- **Booking ID**: Displayed prominently at the top of the page.
- **Room Details**:
    - **Image** of the booked room type.
    - **Check-in and Check-out Dates**.
    - **Total cost** of the complete booking.

2. Collapsible Sections

The following sections are collapsible for better organization:

a. Cost Summary

- Displays a breakdown of costs (Similar to that from the *Trip Itinerary Panel*)
- Includes:
    - Subtotal
    - Taxes and Surcharges
    - Due at Resort
    - Total for the stay

b. Guest Information

- Displays traveler details collected during checkout, including:
    - First & last name.
    - Phone number.
    - Email ID.

c. Billing Information

- Displays billing details collected during checkout, including:
    - Name.
    - Address.
    - City, State, ZIP, Country.
    - Contact details.

d. Payment Information

- Displays **masked** card details for security.
- Shows the last four digits of the card number and expiry date.
- **CVV must not be stored or displayed**.

Production-Ready Requirements for Internet Booking Engine (IBE)

1. Infrastructure Requirements

1.1 Infrastructure-as-Code (IaC)

- Use **Terraform** or **AWS CloudFormation** for provisioning all AWS resources.

1.2 High Availability & Scalability

2. Monitoring, Metrics & Observability

NOTE: *No IAC needed for this. Create a **cloudwatch dashboard** manually.*

2.1 Application and Database Monitoring

- **API Metrics:**
    - Request count, error rates (4xx/5xx), response time (p50, p95, p99).
    - Database query performance.
- **Resource Utilization:**
    - CPU Utilization: Ensure CPU usage stays within optimal thresholds.

    - Memory Usage: Track buffer/cache hit ratios.
    - Disk I/O: Monitor read/write operations on storage disks.
- **Connection Metrics:**
    - Active Connections: Monitor the number of concurrent database connections.
    - Connection Errors: Track failed connection attempts.
- **Replication & Backup Metrics:**
    - Replication Lag: Monitor delays between primary and replica databases.
    - Backup Status: Ensure backups are completed successfully and on schedule.

2.2 Alerts & Incident Management

- **Critical Alerts:**
    - API downtime (>1 min).
    - High error rates (>5% 5xx responses).
    - Database CPU usage > 80%.

3. CI/CD Pipeline & Deployment Process

3.1 CI/CD Pipeline

- Implement pipelines using **AWS CodePipeline** or **GitHub Actions**.
- Stages:
    - **Build → Test → Deploy to Dev → Manual Approval → Deploy to Production**.

3.2 Release Notes & Documentation

- **Release notes** generated with every deployment, documenting features, fixes, and known issues.
- API documentation is **auto-generated** using **Swagger/OpenAPI**.
- Provide an **API console** (e.g., Swagger UI) for testing APIs in real-time.

3.3 Automated Testing & Coverage

- Automated tests covering **unit tests**.
- Test coverage threshold: **≥ 50%**.
- **Regression tests** run before every production release.

4. Performance Testing

4.1 Performance Benchmarks

- APIs must handle **1000+ concurrent users** with minimal degradation.
- Response time targets:

    - **< 500 ms** for 95% of requests.
    - **< 2 seconds** for complex queries.

4.2 APIs to Test for Load & Stress

- **Room Search API:** Test with multiple concurrent queries with various filters.
- **Booking API:** Simulate high booking rates during peak hours.
- **Payment API:** Test secure transactions under load.

4.3 Performance Testing Tools

- Use **JMeter**, **Gatling**, or **k6** for load and stress testing.
- Run performance tests **before every major release**.